

UNIT-I: LINEAR ALGEBRA (Practice Questions)

A. Subspace and Structure

1. For which values of k does the set $W = \{(x, y, z) \in \mathbb{R}^3 : x + ky + z = 0\}$ form a subspace of $\mathbb{R}^3$?
2. Determine whether the set $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 0\}$ is a subspace of $\mathbb{R}^3$.
3. Let $V = \{(x, y, z, w) \in \mathbb{R}^4 : x - y + z - w = 0\}$. Find a parametric representation of V .
4. Determine whether the set of all vectors of the form $(x, x + y, x + 2y)$, where $x, y \in \mathbb{R}$, is a subspace of $\mathbb{R}^3$.

B. Linear Dependence and Independence

5. For which values of a are the vectors $(1, a, 1)$, $(a, 1, 1)$, $(1, 1, a)$ linearly independent?
6. Find all values of k for which the vectors $(1, 2, k)$, $(2, 3, 4)$, $(3, 5, 8)$ are linearly dependent.
7. Determine whether the set $\{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in P_2 .
8. Find a maximal linearly independent subset of the set $\{(1,1,0), (2,3,1), (3,4,1), (1,2,1)\}$.

C. Span, Basis and Dimension

9. Determine whether the set $\{(1,0,1), (0,1,1), (1,1,2)\}$ spans $\mathbb{R}^3$.
10. Find a basis and the dimension of the subspace $V = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + 3z = 0\}$.
11. Express the polynomial $x^2 + 2x + 1$ as a linear combination of $\{x^2 + x, x + 1, 1\}$.
12. Determine whether the set $\{(1,0,1,0), (0,1,0,1), (1,1,1,1)\}$ is a basis for its span in $\mathbb{R}^4$.

D. Linear Transformations

13. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y, y + z, x + z)$. Find:
 - (a) $\text{Ker}(T)$
 - (b) $\text{Range}(T)$
 - (c) Rank and Nullity of T .
14. Determine whether the linear transformation $T(x, y, z) = (x - y, y - z, z - x)$ is one-to-one or onto.
15. Find the standard matrix of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (3x - y, x + 2y)$.

E. Rank of Matrices

16. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + 2y - z, 3x + y + z)$.

- (a) Find the kernel of T in parametric form.
- (b) Find a basis of the range of T and hence determine $\text{rank}(T)$.
- (c) Verify the Rank–Nullity Theorem for T .

17. Let $T : P_2 \rightarrow P_2$ be defined by

$$T(p) = p(t) - p(0).$$

- (a) Find the kernel of T .
- (b) Find the range of T .
- (c) Determine whether T is onto.

18. Find the rank of the matrix for different values of λ :

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & \lambda \end{pmatrix}$$

19. Determine the rank of the matrix $\begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ for the following cases:

- (a) $a = b = c$
- (b) $a + b + c = 0$
- (c) General case

F. System of Linear Equations

20. For which values of k does the system

$$\begin{aligned} x + y + z &= 1 \\ 2x + 3y + kz &= 2 \\ 3x + 4y + 2z &= 3 \end{aligned}$$

have:

- (a) a unique solution,
- (b) infinitely many solutions,
- (c) no solution?

21. Solve the system of equations using Gauss elimination:

$$\begin{aligned} 2x - y + z &= 1 \\ x + 2y + 3z &= 4 \\ 3x + y + 4z &= 5 \end{aligned}$$

22. Check the consistency of the system:

$$\begin{aligned} x + 2y + 3z &= 1 \\ 2x + 4y + 6z &= 3 \\ 3x + 6y + 9z &= k \end{aligned}$$